

PROJECT_CONTEXT.md

CommercePilot

Version: 1.0

Document Type: AI Context Engineering

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1. Project Identity

Project Name

CommercePilot

Tagline

AI Commerce Automation Platform for Existing E-commerce Businesses

Project Type

Multi-tenant SaaS Platform

AI Automation Platform

Plugin-based Commerce Integration Platform

Final Year Computer Science Project

Potential Commercial Startup

2. Project Vision

CommercePilot is an AI-powered commerce automation platform designed to integrate with existing e-commerce systems rather than replace them.

The platform connects to an existing online store (WooCommerce, Shopify, or custom APIs), monitors customer communication channels such as WhatsApp, understands customer intent using Large Language Models, automates repetitive business processes, and assists business owners with intelligent order processing.

CommercePilot is not a chatbot.

CommercePilot is an AI Employee.

3. Mission

Reduce manual work performed by online business owners through AI-driven workflow automation while maintaining full human control over critical business decisions.

4. Problem Statement

Many SMEs receive orders through WhatsApp instead of directly through their websites.

Business owners manually:

- Read customer messages
- Extract order details
- Check inventory
- Create orders
- Send quotations
- Reply to customers
- Update inventory
- Notify staff

This process is:

- Slow
- Error-prone
- Difficult to scale
- Expensive

CommercePilot automates these repetitive tasks while allowing the owner to review and approve AI decisions when required.

5. Target Users

Primary Users

- Small Business Owners
- Medium-sized E-commerce Stores
- Textile Businesses
- Fashion Stores
- Electronics Stores
- Cosmetics Stores
- Gift Shops

Secondary Users

- Sales Staff
 - Customer Support Teams
 - Warehouse Staff
-

6. Core Philosophy

The system must NEVER replace the business owner.

The system assists the owner.

Human approval always has higher authority than AI.

AI recommendations are suggestions.

The owner always has the final decision.

7. Product Principles

1. AI First

Every repetitive task should be evaluated for automation.

1. Human in the Loop

Critical business decisions require owner approval unless explicitly configured otherwise.

1. Existing Systems First

CommercePilot never replaces existing e-commerce systems.

CommercePilot extends them.

1. Multi-Tenant by Design

Every customer business is isolated.

No tenant data may ever be accessible by another tenant.

Tenant isolation is mandatory.

1. API First

Every feature must be accessible through APIs.

Internal communication should follow API contracts.

1. Security First

Security is designed into the architecture.

Never added later.

1. Production Ready

Every implementation must be scalable.

Temporary code is prohibited.

8. Primary Features

AI WhatsApp Order Processing

AI Order Extraction

Order Approval Dashboard

WooCommerce Integration

Shopify Integration (Future)

Customer Conversation Management

AI Confidence Scoring

Business Analytics

Workflow Automation

Email Notifications

Inventory Validation

RAG Product Search

Owner Approval Workflow

9. MVP Scope

Included

- ✓ WhatsApp Integration
- ✓ AI Order Extraction
- ✓ WooCommerce Integration
- ✓ Owner Dashboard
- ✓ Email Notifications
- ✓ Human Approval
- ✓ PostgreSQL Database
- ✓ Multi-Tenant Architecture
- ✓ Authentication

Excluded

- ✗ Shopify
- ✗ Facebook Messenger
- ✗ Instagram
- ✗ Mobile App
- ✗ Sinhala NLP
- ✗ Voice Order Processing

These belong to future releases.

10. Success Criteria

The MVP is considered successful when:

A customer sends a WhatsApp message.

↓

AI understands the order.

↓

AI extracts structured information.

↓

Inventory is validated.

↓

The owner receives a notification.

↓

The owner approves the order.

↓

The order is automatically created inside WooCommerce.

↓

The customer receives confirmation.

No manual data entry should be required.

11. Non-Functional Requirements

High Availability

Scalable

Secure

Maintainable

Modular

Cloud Ready

API Driven

Production Quality

Well Tested

Well Documented

12. Architecture Principles

Clean Architecture

Domain Driven Design

SOLID Principles

Repository Pattern

Dependency Injection

Event Driven Processing

Queue Based Background Jobs

Multi-Tenant Isolation

REST APIs

Webhook Driven Integrations

13. AI Principles

The AI does not make business decisions.

The AI recommends.

The owner approves.

The AI must never hallucinate products.

The AI must use Retrieval-Augmented Generation (RAG) against the tenant's product catalog before making recommendations.

Confidence scores must determine whether human review is required.

14. Technology Stack

Frontend

Next.js

Backend

NestJS

Database

PostgreSQL

ORM

Prisma

Authentication

Supabase Auth

AI

Gemini Flash (MVP)

Future: Claude / OpenAI

Automation

BullMQ

Redis

Hosting

Render

Frontend Hosting

Vercel

Emails

Resend

Storage

Supabase Storage

Vector Search

pgvector

15. Development Rules

No shortcuts.

No duplicate logic.

No hardcoded secrets.

No insecure code.

No placeholder implementations.

Every feature must include:

Documentation

Tests

Error handling

Logging

Validation

Security review

16. Long-Term Vision

CommercePilot becomes a plug-and-play AI automation platform that any e-commerce business can install in less than 30 minutes.

The platform evolves into an AI operations layer capable of automating customer communication, order processing, business analytics, inventory intelligence, and workflow orchestration across multiple commerce platforms.